



1. Overview

Benjamin English is a web app where English learners look up a word, read its definition and examples, listen to its pronunciation, and save it to a personal vocabulary list. A review mode and a progress page show which words are still new and which are learned.

Problem: students use a dictionary, a translator, and a notebook to study one word.

Motivation: I teach English and want a simple, free tool my students can use to study on their own.

2. Target Audience

Spanish-speaking students learning English at levels A1–B2, especially students of Benjamin Academy. Teachers can also recommend it. No account is needed; data stays on the user's device.

3. Major Functions

Function	Description
Word search	Search any English word, with a clear message if it is not found.
Definitions and examples	Shows part of speech, definitions, and example sentences.
Pronunciation	Shows the phonetic spelling and plays the audio.
Accent explorer	Shows English-speaking countries with flag and region to compare accents (US, UK, Australia).
Save word	Saves the word with a level (A1–B2) and an optional note in Spanish.
Vocabulary list	Lists saved words, filters by level and status (new, learning, learned), and allows delete.

Flashcard review	Flip cards to practice, and mark each word as known or not.
Progress page	Shows total words, words per status, and a daily study streak.

4. Wireframes

1. Search

Mobile

Logo

Search bar

Word · Phonetic · Audio
serendipity

Definition and example

Save word

Accents

Menu: Search · Vocabulary
· Review · Progress

Desktop

Logo · Search · Vocabulary · Review · Progress

Search bar

Word · Phonetic · Audio
Definition and example

Save word

Accents: US · UK · AU

2. Vocabulary and progress

Mobile

Logo

Total words · Learned ·
Streak

Filters

serendipity · B2 · learning

library · A1 · learned

Start review

Menu

Desktop

Logo · Search · Vocabulary · Review · Progress

Total words

Learned

Streak

Filters · Start review

serendipity · B2 · learning

library · A1 · learned

3. Flashcard review

Mobile

Desktop

Logo

Card 3 of 10

Flashcard
serendipity

Didn't know

Knew it

Menu

Logo · Search · Vocabulary · Review · Progress

Card 3 of 10

Flashcard
serendipity

Didn't know

Knew it

5. External Data

- **Free Dictionary API** (`api.dictionaryapi.dev`): definitions, phonetics, audio, examples.
- **REST Countries API** (`restcountries.com`): flags, regions, and languages of English-speaking countries.
- **Stored in localStorage**: saved words (word, definition, level, note, status) and progress (review results, streak).

6. Module List

Module	Responsibility
<code>dictionaryApi.js</code>	Fetches word data from the Free Dictionary API.
<code>countriesApi.js</code>	Fetches countries from REST Countries.
<code>storage.js</code>	Saves and reads words and progress in localStorage.
<code>vocabulary.js</code>	Saved word list, filters, and status changes.
<code>review.js</code>	Flashcards and progress statistics.
<code>main.js</code>	Search view, navigation, and app start-up.

7. Graphic Identity



- **Typography:** Ubuntu (humanist sans-serif), bold for headings and words, regular for body text.
- **Styling:** white screens, gray rounded borders and pill-shaped buttons. The gradient is used only for key actions (save, play, progress bar) and flashcard highlights; green marks learned words.
- **Inspiration:** the visual style is inspired by Instagram: Most of my students spend their time on that app. I want them to feel comfortable and engaged while using my app.
- **Icon:** a rounded square with the gradient, a white speech bubble, and "Aa" in pink (shown above).

8. Timeline

Week	Deliverables
5	Project setup, responsive layout and navigation, dictionary API, search view with definitions and audio.
6	localStorage, save word, vocabulary list with filters, REST Countries accent explorer.
7	Flashcard review, progress page, testing, bug fixes, and deployment.

9. Project Planning

Trello board: <https://trello.com/b/JZ3mnbhu/benjamin-english-wdd-330-final-project>

10. Challenges

- **Incomplete API data:** some words may have no audio or examples, or may not be found. The app needs clear fallbacks.
- **Time:** eight functions in three weeks, 4 functions per week, last week for testing and deployment.

- **Responsive design:** the app must work well on phones and I want to make it installable as a PWA